Barbershop Scheduling Analysis

Customer flow, service duration and capacity for a single barber over four weeks.

About the data:

The dataset is simulated (237 visits, 4-30 August 2026), so the findings demonstrate the method rather than describe a real shop. Every figure in this report is printed or plotted in the accompanying notebook; the notebook step is named under each finding.

Overview

Question: Across four weeks, how does the barber's workload vary by day and time, and where is the schedule over or underbooked?

237	**30.6 min**	**6.7 min**	**63%**
visits across 24 days	mean service duration	mean wait (maximum 17)	average utilisation of a 480-minute day

Key Statistics:

- **Visits by day**. Saturday (52), Friday (47) and Sunday (40) have the most visits; Wednesday (33) and Tuesday (29) have the fewest.
- **Utilisation by day**. Average utilisation of a 480-minute day runs from 48.1% (Tuesday) to 80.7% (Saturday). No day exceeds 100%; the highest single day is Friday 7 Aug at 90.6%.
- **Utilisation within the day**. The highest block average is Friday evening (98%) and the lowest are Tuesday and Wednesday mornings (17%). 10 of the 72 day-and-block slots had more booked minutes than their capacity, the highest being 133% (Friday 28 Aug evening).
- **Wait time**. The evening block has the highest average wait (7.81 min, against 5.83 in the morning). The highest day-and-block averages are Saturday and Sunday evenings (9.9 min each).
- **Service duration and weekly totals**. Colour/Chemical has the longest mean duration (60.6 min) and Beard Trim the shortest (15.2 min). Weekly visits were 60, 57, 62 and 58.

Data and Methodology

Each row is one client visit with its date, day of week, time block, service type, service duration, wait time and booking type. There are no client names or clock times, nor are there any prices of the services provided. Through validation, the notebook found 0 missing values, with no spelling or capitalisation variants in the four category columns, and visits on 24 distinct dates between 4 and 30 August with no Monday visits (days-off). Two rows repeat an earlier row exactly; as the data has no visit ID column they were left in.

Assumptions:

Capacity Assumptions: an 8-hour (480-minute) day, and block capacities of 180 (Morning), 240 (Afternoon) and 180 (Evening) minutes. Utilisation is booked service minutes divided by capacity, and idle minutes are capacity minus booked minutes, so any break or admin time counts as idle.

Key findings

Visits

Notebook source: Step 4 (visits per weekday per week, totals, average service hours per day)

Saturday has the most visits in weeks 1 to 3 and ties with Friday in week 4 (12 each). Tuesday has the fewest in weeks 1, 2 and 4 and ties with Wednesday in week 3 (8 each). Average booked service time per day (total service hours for that weekday divided by 4 weeks) is 6.5 h on Saturday, 6.2 h on Friday, 5.1 h on Sunday, 4.6 h on Thursday, 4.0 h on Wednesday and 3.8 h on Tuesday.

This tells us that Saturday and Friday are the busiest days, and Saturday never has fewer visits than Friday (it is ahead in weeks 1 to 3 and level in week 4). Tuesday and Wednesday are the quietest. Workload clearly differs by weekday, which is enough to justify looking at each day separately.

From looking at figure 2, it also shows how the day gets progressively busier with the evening time block (4-7) having the most visits.



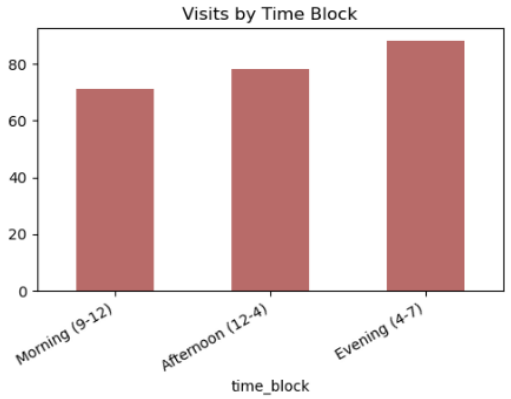


Figure 1. Total visits by day of week (left). Figure 2. Total visits by time block (right).

Capacity and idle time

Notebook source: Step 6 (utilisation table, summary prints, per-slot table, block heatmap)

Utilisation compares booked minutes with available minutes, assuming a 480-minute day. Across the four weeks the barber is booked for 63% of that time, but the average hides a wide spread between days: 48.1% on Tuesday against 80.7% on Saturday. The gap matters more than the average, because it shows the same working day is used very differently across the week.

Idle time is concentrated rather than spread evenly. Tuesday to Thursday hold 46 of the 71 idle hours, so these days are where there is most unused time to recover.

The block view shows what the daily figure hides. No full day exceeds 100%, yet Friday evening averages 98% and 10 of 72 day-and-block slots had more booked minutes than their capacity (the highest was 133%). Meanwhile Tuesday and Wednesday mornings average 17%. The busiest periods are short-lived inside otherwise manageable days: evenings on Tuesday to Friday, mornings on Saturday and Sunday. So demand is better described by time of day than by day alone.

These figures show where to test scheduling changes, such as moving work from the tightest blocks towards the emptiest or adjusting opening hours. They cannot show whether clients would shift, or whether demand was turned away when blocks were full, because the data only records visits that happened.

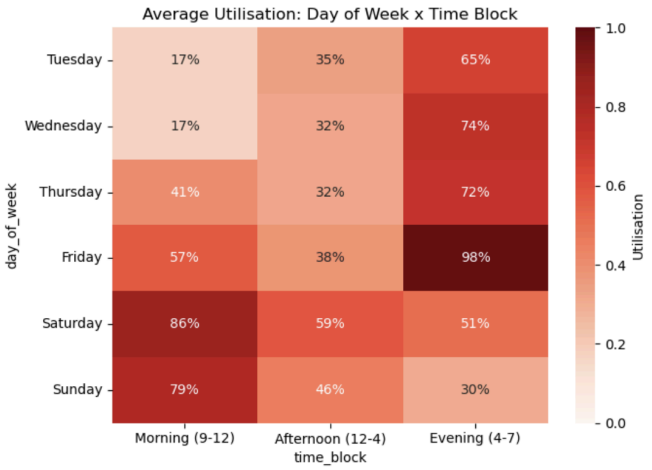


Figure 3. Average utilisation by day and time block (booked minutes averaged over 4 weeks as a share of block capacity).

Wait times

Notebook source: Step 5 (wait by day, by time block, and day x time block heatmap)

Waits are short overall: 6.7 minutes on average, with a maximum of 17. Averages vary by only about two minutes across days (5.79 on Wednesday to 7.75 on Saturday) and across time blocks (5.83 in the morning to 7.81 in the evening). The larger differences appear when day and time block are combined: Saturday and Sunday evenings average 9.9 minutes, against 3.2 on Wednesday morning.

Wait does not simply track how full a block is. Weekend evenings have the longest waits but low utilisation (51% and 30%), while Friday evening, the fullest block at 98%, averages 6.1 minutes. The data has no arrival times, so it cannot show what drives these waits. That makes them a question to investigate.

Service mix and duration

Notebook source: Steps 2 and 5 (category counts, duration by service type, booking type)

Haircut is the most frequent service (110 of 237 visits), followed by Haircut + Beard (46) and Beard Trim (42). Average duration differs widely by service, from 15.2 minutes for a Beard Trim to 60.6 for Colour/Chemical, which is rare (8 visits) but takes about an hour each. Durations within each service are fairly consistent: the standard deviation ranges from 2.4 minutes (Beard Trim) to 8.2 (Colour/Chemical), much smaller than the gaps between services. Service type therefore explains most of the variation in how long a visit takes.

Walk-ins (119) and appointments (118) are split almost evenly, with walk-ins averaging 2.6 minutes longer (31.9 against 29.3). The analysis does not test whether this is more than normal variation, or whether it reflects a different mix of services rather than booking type, so it is not treated as a finding.

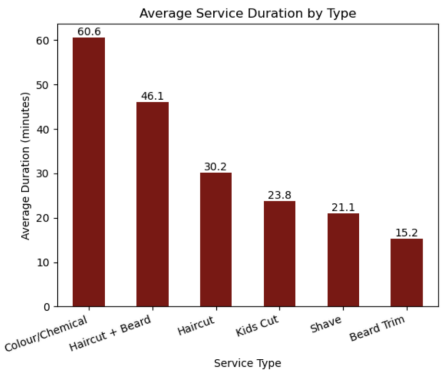


Figure 4. Mean service duration by type.

Trend over the four weeks

Notebook source: Step 7 (weekly table, week x service type table)

Weekly visits were 60, 57, 62 and 58, and total service minutes were 1797, 1888, 1751 and 1823. Neither rises or falls steadily across the weeks. Average service duration moved more (29.95, 33.12, 28.24 and 31.43 minutes), with week 2 the highest. Week 2 also had the most Colour/Chemical services (5, against 1 in each other week), the longest service type, which is one possible explanation the notebook does not test. Visits on individual weekdays are also similar from week to week (for example Saturday 14, 12, 14, 12).

Four weeks is too short to confirm a trend or to separate a trend from normal week-to-week variation. A longer record would be needed before using this to forecast demand.

Higher and Lower Utilisation

This section groups the notebook's figures by how much of the available time was booked and notes what the pattern suggests.

Lower utilisation

- Tuesday and Wednesday average 48.1% and 49.9% of a 480-minute day. Tuesday to Thursday hold 46 of the 71 idle hours, so unused time sits early in the week.
- Within those days the slack is in the daytime. Tuesday to Thursday mornings and afternoons run at 17% to 41% of block capacity, while their evenings run at 65% to 74%.

Higher utilisation

- Friday evening averages 98%, Saturday morning 86% and Sunday morning 79%. The busy window moves from the evening on weekdays to the morning at weekends, and it is a window, not a whole day: Friday afternoon is only 38%.
- No full day exceeds 100% (the busiest averages 80.7%), yet 10 of 72 day-and-block slots did, up to 133% (Friday 28 Aug evening). Pressure shows at block level and is hidden by the daily figure.

Waits and durations

- The longest waits are on weekend evenings (9.9 min), which are not the fullest blocks (Saturday 51%, Sunday 30%). Friday evening, the fullest at 98%, averages 6.1 min. Wait does not simply follow utilisation, and the data cannot show why.
- Colour/Chemical (60.6 min) and Beard Trim (15.2 min) differ widely, and utilisation counts minutes, not visits, so the mix of services in a block affects how full it looks.

What this tells us: the same days contain both unused time and tight blocks. Friday alone runs from 38% to 98% across its blocks, wider than the gap between the quietest and busiest day averages (48.1% to 80.7%). When demand falls matters more than which day it falls on.

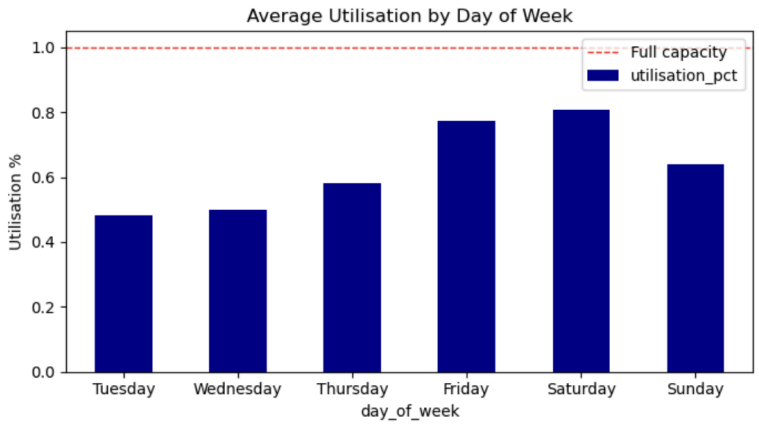


Figure 5. Average utilisation by day of week, in percentage.

Evaluation

This analysis addresses one central question: across four weeks, how does the barber's workload vary by day and time, and where is the schedule over- or under-booked?

To answer it, the question was broken into five smaller sub-questions, each covering one part of the picture. Together their answers build up the overall verdict. The sub-questions were:

1. Demand Pattern - Which days are busiest/quietest? Does demand concentrate in a particular time block?
2. Service Mix & Duration Trends - Which services does the barber do most? Which take the longest? Is there a difference in duration between walk-ins and appointments?
3. Wait Time - When do clients wait longest? Does wait time correlate with how busy that day/block is?
4. Capacity & Idle Time - Roughly how many service minutes is the barber delivering per day vs. a fixed shift length? Which days/blocks show the most idle capacity vs. overload?
5. Trend over the 4 weeks - Is the total weekly volume rising or falling?

Demand Pattern:

- Days: Saturday is busiest with 52 visits (Friday 47, Sunday 40). Tuesday is quietest with 29 (Wednesday 33).
- Time block: demand does concentrate, but the pattern depends on the day. On Tuesday to Friday the evening block has the highest utilisation (65%, 74%, 72%, 98%), and on Saturday and Sunday the morning does (86%, 79%).

Service Mix & Duration Trends:

- Most frequent: Haircut, with 110 of 237 visits.
- Longest: Colour/Chemical at 60.6 min on average, then Haircut + Beard at 46.1. The shortest is Beard Trim at 15.2 min.
- Walk-ins vs appointments: walk-ins average 31.9 min and appointments 29.3, a 2.6 min difference. -
- "Rushed vs Unhurried" cannot be answered with insufficient data.

Wait Time:

- Longest waits: Saturday and Sunday evenings at 9.9 min each. By block, Evening is highest at 7.81 min (Morning 5.83), and by day, Saturday is highest at 7.75 min.
- The heatmaps suggest wait doesn't simply follow utilisation: Friday evening is the fullest block (98%) with a 6.1 min wait, while Sunday evening is only 30% utilised yet waits 9.9 min.

Capacity & Idle Time:

- Service minutes per day against 480: from 230.8 min on Tuesday to 387.5 on Saturday, with 63% overall utilisation.
- Idle: 71 hours in total, 46 of them on Tuesday to Thursday. The most idle blocks are Tuesday and Wednesday mornings at 17% utilisation.
- Overload: no day exceeds 100% (the highest is 90.6%, Friday 7 Aug), but 10 of 72 day-and-block slots did, up to 133% (Friday 28 Aug evening).

Trend over the 4 weeks:

- Weekly volume: 60, 57, 62 and 58 visits, so there is no steady rise or fall. Four weeks is too few to call it a trend.
- No day exceeds 100% in any average. Visits per weekday are similar week to week (Saturday 14, 12, 14, 12; Tuesday 7, 7, 8, 7). The notebook doesn't show utilisation per day per week, so this part of the question may need to be explored further.

Limitations

The data is simulated and covers four weeks, so each weekday has only four observations (Table 1). Utilisation depends on the 480-minute and block-capacity assumptions, and breaks count as idle time. The data has time blocks but no arrival or start times, so the order of visits within a block cannot be seen. Wait time is recorded per visit and does not show what caused a queue. The notebook reports averages and counts only; it contains no statistical tests.